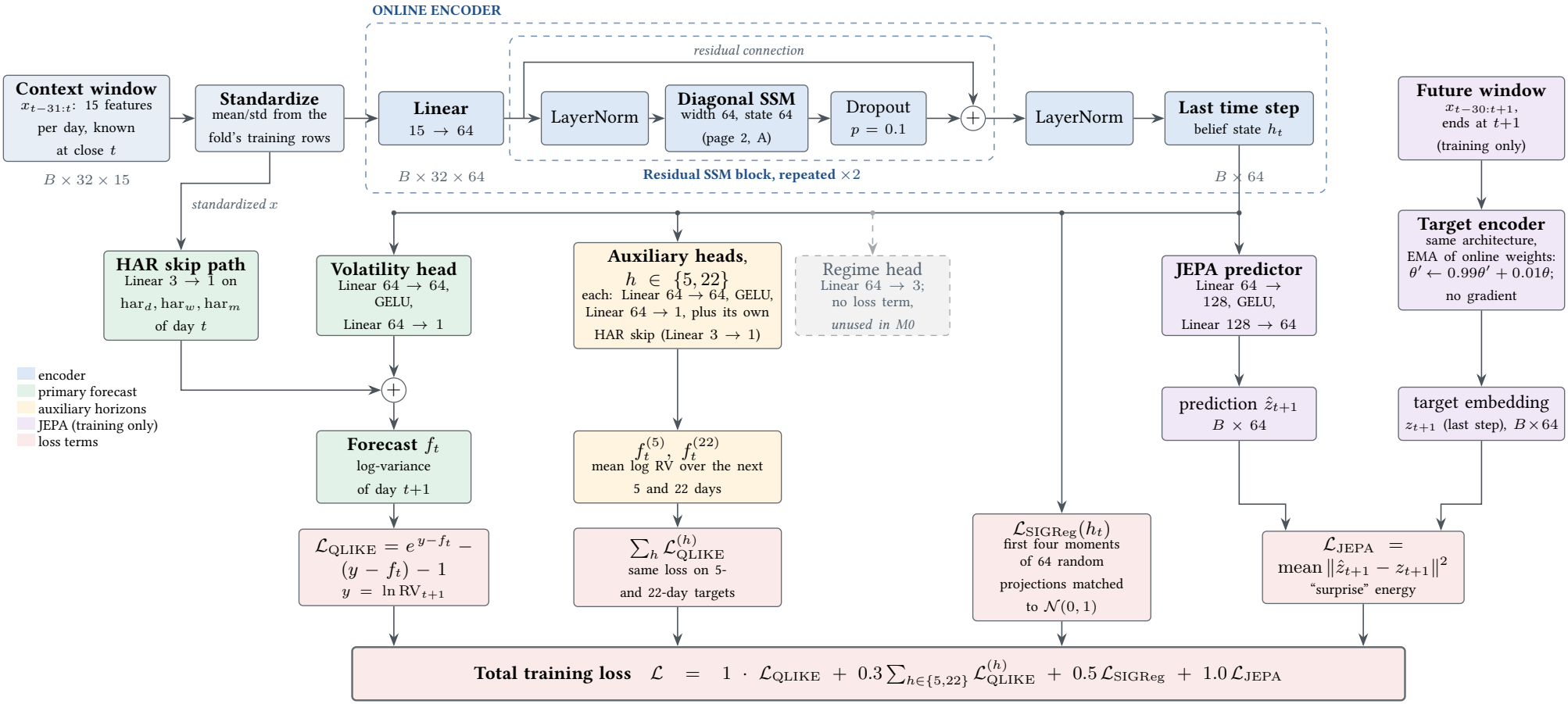
Meridian M0 — network architecture as used in the audited benchmark

Source: `meridian/model.py` with configuration `variant_cfg("M0")` in `scripts/benchmark_vol.py`. Shapes: B = batch, 32 days, 15 features, width 64, SSM state 64.

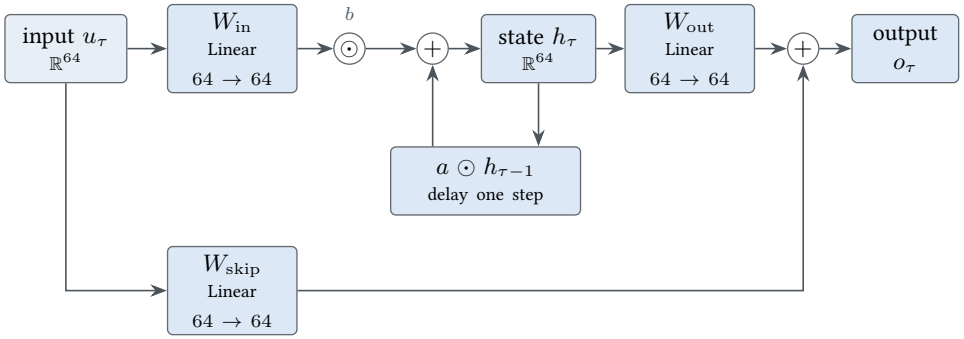


What leaves the network. Only f_t is used as the volatility forecast ($f_t^{(5)}$, $f_t^{(22)}$ only for the horizon tables). The JEPA branch, the target encoder and SIGReg act only during training. **Ensemble.** 5 independently seeded copies are trained in every walk-forward fold, and the reported forecast is the mean of their f_t. **Initialization.** The HAR skip paths start at the fold's HAR regression and the last layers of the heads start at zero, so f_t equals the HAR-RV forecast before training (page 2, B).

Inside the network: the SSM layer, the HAR-anchored start, the training loop

Source: `DiagonalSSM`, `Encoder`, `Meridian.init_har_anchor` in `meridian/model.py`; `train_meridian`, `run_meridian_fold` in `scripts/benchmark_vol.py`.

A. Diagonal linear state-space layer (one per residual block)



$$h_\tau = a \odot h_{\tau-1} + b \odot W_{\text{in}} u_\tau, \qquad o_\tau = W_{\text{out}} h_\tau + W_{\text{skip}} u_\tau, \qquad h_0 = 0, \quad \tau = 1, \dots, 32.$$

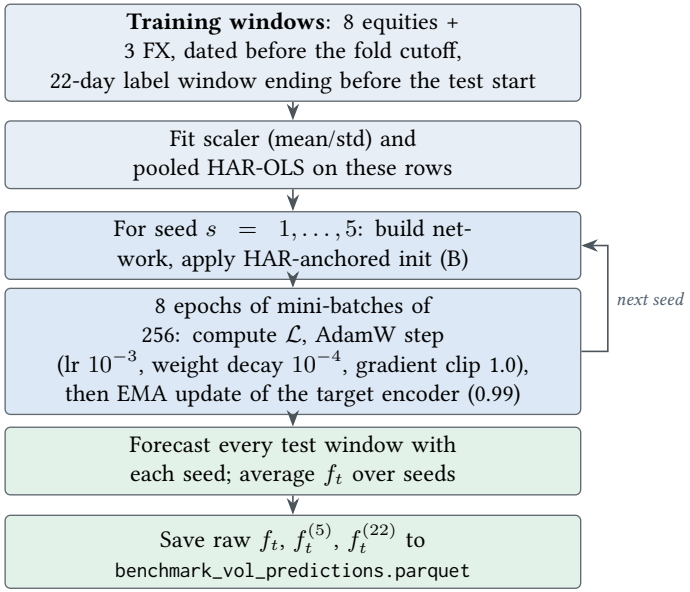
$a = \text{sigmoid}(\alpha)$ is a learned decay per state channel, initialized with $\alpha = \text{linspace}(1, 4)$, i.e. $a \in (0.73,\ 0.98)$: each channel is a stable exponential memory with its own time scale. b is a learned input gain, initialized to 0.5. The recurrence is scanned sequentially over the 32 days. In the encoder each layer is wrapped as $x \leftarrow x + \text{Dropout}(\text{SSM}(\text{LayerNorm}(x)))$; two blocks are stacked and followed by a final LayerNorm.

B. HAR-anchored initialization (HARNet-style), every fold, every seed

1. Fit pooled HAR by OLS on the fold's core-asset training rows: $\ln \text{RV}_{t+1} = \beta_0 + \beta_d \,\text{har}_d + \beta_w \,\text{har}_w + \beta_m \,\text{har}_m$ (and the same regression for the 5- and 22-day targets).
2. Set the HAR skip path to weights $w = \beta_{1:3} \odot \sigma_{\text{har}}$ and bias $\beta_0 + \beta_{1:3}^\top \mu_{\text{har}}$, where $\mu_{\text{har}}, \sigma_{\text{har}}$ are the scaler's statistics, so the skip path reproduces HAR on standardized inputs.
3. Set the final linear layer of the volatility head and of both auxiliary heads to zero.

Consequence: before the first gradient step, f_t is exactly the HAR-RV forecast. Training only learns corrections to HAR.

C. Training and forecasting in one walk-forward fold

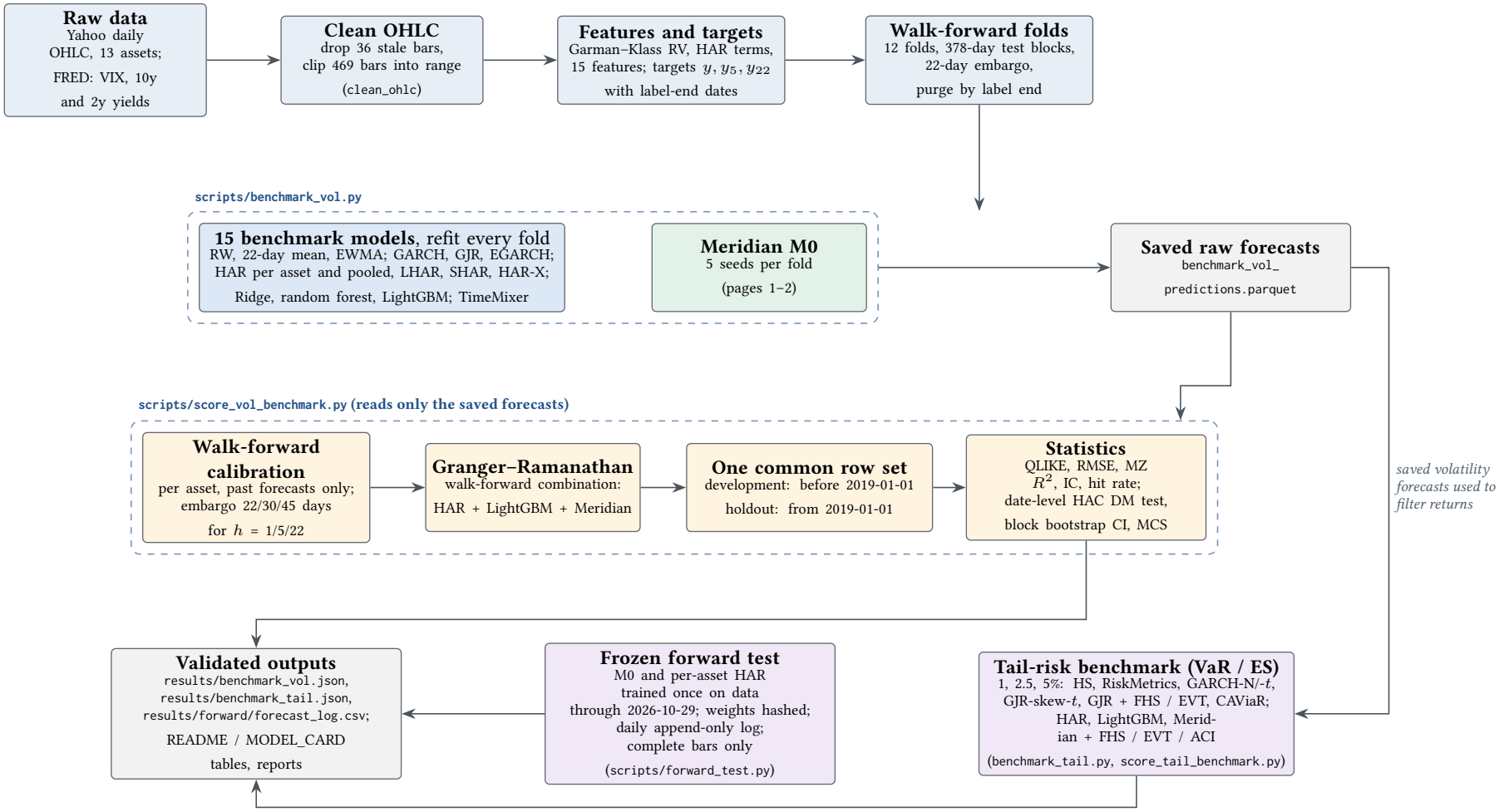


D. M0 configuration (exact values from the code)

Input window / features	32 days / 15
Model width / SSM state / SSM blocks	64 / 64 / 2
Dropout	0.1
Loss weights: QLIKE, auxiliary, SIGReg, JEPA	1, 0.3, 0.5, 1.0
EMA rate of the target encoder	0.99
Optimizer	AdamW, lr 10^{-3}, wd 10^{-4}
Batch / epochs / gradient clip	256 / 8 / 1.0
Seeds per fold	5

End-to-end system: data, walk-forward benchmark, scoring, tail risk, forward test

Each block names the script that implements it. Forecasts are saved before any metric is computed, so every reported number is recomputed from saved files.



Leakage controls, enforced in code and checked by 11 automated tests. Every scaler, regression and network is fit on rows dated before the fold cutoff whose label window ends before the test start; calibration and combination use only past forecasts; a tail forecast for day $t+1$ uses standardized returns of earlier days only; all model choices use the development period only; the forward test logs a forecast before its target day exists.